College Name	**University of Hertfordshire** **SEGi College Subang Jaya**		
Programme Name	**BACHELOR OF CYBERSECURITY AND NETWORKS**		
Module Name	CYBER SECURITY AND NETWORKS PROJECT	Module Code	6COM2019
		Semester	**SEP 2025**
Module Leader	MS ANIZAH ABU BAKAR	Assessment Type	PROGRESS REPORT
Lecturer Name	MS NUR DIANA MADINAH		
Student's declaration	I hereby certify that this assignment is my own work and where materials have been used from other resources, they have been properly acknowledged. I also understand I will face the possibility of failing the module if the content of this assignment is plagiarized.		

No.	Name	Student ID	Signature / Initial
1	TAN JIAJIAN	SCSJ1901781	TAN

Release Date		Submission Due Date	28/11/2025	Marks obtained:
Date Received		Student's work assessed by / date		

Module Leader's Feedback.

Contents

1. Introduction

This project focuses on developing a Secure Chat Application that supports real-time communication protected by end-to-end encryption (E2EE). The goal is to ensure that messages can only be accessed by the intended sender and receiver by using strong cryptographic algorithms such as RSA, AES and Diffie-Hellman. The system aims to solve the growing security challenges in digital communication, particularly the risks of unauthorized access, data interception and man-in-the-middle (MITM) attacks (Mallik. A, 2019).

The main problem of this project is the lack of fully secure messaging in many existing chat applications. Some platforms still depend on centralized servers or incomplete encryption methods, making sensitive user information vulnerable to cyberattacks (Nalla, 2024). With communication becoming increasingly digital, there is an urgent need for a secure, reliable and privacy focused chat system (Dashtinejad, 2015).

At this stage, the project is still in the planning phase. The research on existing secure messaging system has been completed and the project requirement has been identified. The project backlog has been drafted, outlining key features such as encryption modules, user authentication and real-time messaging. The next steps involve finalizing system design, selecting suitable development tools and preparing for implementation in the upcoming sprints.

2. Project Progress

So far, the project is progressing according to the initial plan, with all early activities for the planning phase successfully completed. The first major task accomplished was the research and analysis of existing secure messaging applications such as WhatsApp, Telegram, Messenger and others that use encryption. This research helped identify their strengths and weaknesses and limitations, especially regarding end-to-end encryption, secure key exchange and user privacy. Findings from this research were used to refine the problem statement and confirm the need for a more transparent and secure communication system.

Another key milestone completed is the comparison of different encryption methods such as RSA, AES and Diffie-Hellman to understand how they can be combined for an effective hybrid cryptographic model. This analysis provided the foundation for designing the core security features of the project.

In terms of planning and design preparation, the project now has a clear set of requirements. A product backlog has been drafted, listing the essential features such as encryption modules, real-time messaging functions, user authentication and secure key exchange. Tasks have been organized and prioritized according to the Agile methodology, preparing the project for sprint-based development.

Additionally, the project proposal was completed and successfully presented. Feedback from the panel has been reviewed into the refined scope and problem definition. Now, coding, testing and system implementation have not started yet. The next step will focus on finalizing the system design, selecting the development tools and preparing for the first sprint of implementation in the next phase.

3. Challenges and Adjustments

During the early planning phase of the Secure Chat Application project, several challenges were encountered, mainly related to technical understanding and project scope refinement. One of the challenges was the complexity of comparing different encryption algorithms such as RSA, AES and Diffie-Hellman. Each algorithm has different strengths, performance levels and intended uses, making it difficult to determine the most suitable combination for a real-time chat application. To solve this, additional research was conducted through academic papers and technical documentation, which helped clarify how hybrid encryption models are commonly implemented. (Davis, T, 2003; Abdullah & Muhamad, 2017; Biswas, 2020)

Another challenge is to understand the limitations of existing chat applications and identify the shortcomings that this project can effectively solve. Many modern applications already provide strong encryption, so it was necessary to clarify the issues more clearly. After reviewing feedback from the panel, the project scope was adjusted to focus more on demonstrating how encryption works rather than competing with full features commercial instant messaging platforms.

Managing project time and setting realistic deliverables also presented early challenges. The original plan included too many features such as file sharing and group chats, which were difficult to implement within the project deadline. Therefore, the project scope was narrowed down to include only essential features such as user authentication, real-time messaging and end-to-end encryption.

Finally, use of agile methodology to project development requires reorganizing tasks into smaller and more manageable iteration cycles. The product backlog has also been revised to prioritize core security features, followed by UI design and secondary functionalities. These adjustments have allowed the project to maintain a clear direction while staying flexible for future changes during development.

4. Updated Project Plan

At this stage of the project, no major changes have been made to the original project plan. After reviewing the proposal feedback and evaluating the overall timeline, the existing plan is still suitable and achievable within the final year project duration. The scope, methodology and scheduled phases remain the same as previously outlined.

The project is currently in the design phase and all activities are progressing according to the initial timeline. The task scheduled for later phases such as development, testing, deployment, review and documentation continue to follow the same sequence and deadlines established in the original plan. Since no delays or new requirements have been introduced, there is no need to modify the Gantt chart or restructure the timeline.

The Agile methodology will continue to guide the project with development and testing carried out iteratively in upcoming sprints as originally planned. Overall, the project remains on track and the previously defined schedule still supports the successful completion of all required deliverables without requiring any adjustments.

5. Next Action Plan

With the planning and design stages now completed, the project will move into the development phase which is structured into four main stages. First phase will focus on building the user authentication system and the basic chat interface. This includes creating the login and registration functions, setting up user identify handling and developing the main chat window where users can type and view messages. This step ensures the foundation of the application is ready before implementing communication features.

The second phase will involve implementing the message sending and receiving functions. At this stage, the system will be able to establish real-time communication between two users in a client-server environment. Basic message transfer will be tested to confirm that the communication pipeline is stable before encryption is applied.

Next, the third phase will integrate the core security features of the project is RSA and AES encryption along with the Diffie-Hellman key exchange mechanism. This step ensures that all messages transmitted are fully protected through end-to-end encryption. Testing will be performed throughout this phase to verify that encrypted messages can be transmitted, received and correctly decrypted.

The fourth development phase will involve optimizing performance, fixing bugs and enhancing the user interface. This includes improving message speed, fixing any errors found during testing, refining the UI layout and ensuring smooth user interaction.

After these four phases, the project will proceed to full system testing, deployment and preparation of the final documentation and presentation. This structured plan ensures steady progress toward delivering a secure and functional chat application.

6. References

1. Davis, T. (2003). RSA Encryption. http://geometer.org/mathcircles/RSA.pdf

2. Abdullah, A., & Muhamad, A. (2017). Advanced Encryption Standard (AES) Algorithm to Encrypt and Decrypt Data Advanced Encryption Standard (AES) Algorithm to Encrypt and Decrypt Data. https://www.researchgate.net/profile/Ako-Abdullah/publication/317615794_Advanced_Encryption_Standard_AES_Algorithm_to_Encrypt_and_Decrypt_Data/links/59437cd8a6fdccb93ab28a48/Advanced-Encryption-Standard-AES-Algorithm-to-Encrypt-and-Decrypt-Data.pdf

3. Biswas, S. (2020, August 21). Diffie-Hellman key exchange in End-to-End Encryption (E2EE). Medium. Retrieved from https://shubhomoybiswas.medium.com/diffie-hellman-key-exchange-in-end-to-end-encryption-e2ee-2366e056661

4. Mallik, A. (2019). MAN-IN-THE-MIDDLE-ATTACK: UNDERSTANDING IN SIMPLE WORDS. Cyberspace: Jurnal Pendidikan Teknologi Informasi, 2(2), 109. https://doi.org/10.22373/cj.v2i2.3453

5. Dashtinejad, P. (2015). Security System for Mobile Messaging Applications [Master's thesis, KTH Royal Institute of Technology]. DiVA portal. Retrieved from https://www.diva-portal.org/smash/get/diva2:813095/fulltext01.pdf

6. Nalla, K. K. (2024). Securing Chat applications: Strategies for end-to-end encryption and cloud data protection. World Journal of Advanced Engineering Technology and Sciences, 13(02), 528–540. https://wjaets.com/sites/default/files/WJAETS-2024-0634.pdf